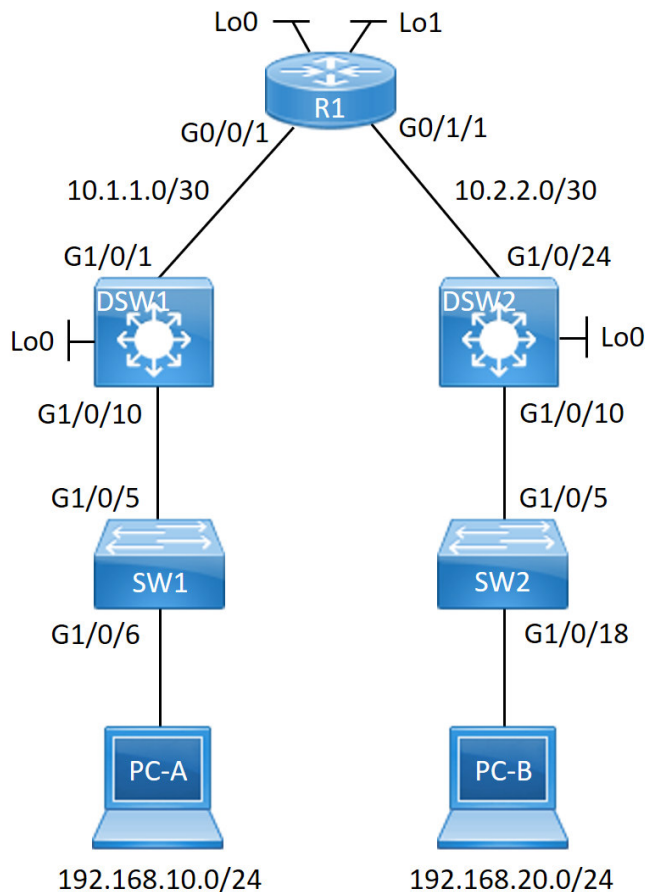


Lab – Configuring and Verifying Extended ACLs

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
DSW1	VLAN 1	192.168.10.1	255.255.255.0	N/A
	Lo0	192.168.30.1	255.255.255.0	N/A
	G1/0/1	10.1.1.1	255.255.255.252	N/A
R1	G0/0/1	10.1.1.2	255.255.255.252	N/A
	Lo0	209.165.200.225	255.255.255.224	N/A
	Lo1	209.165.201.1	255.255.255.224	N/A
	G0/1/1	10.2.2.2	255.255.255.252	N/A
DSW2	G1/0/24	10.2.2.1	255.255.255.252	N/A
	VLAN 1	192.168.20.1	255.255.255.0	N/A
	Lo0	192.168.40.1	255.255.255.0	N/A
PC-A	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-B	NIC	192.168.20.3	255.255.255.0	192.168.20.1

Objectives

Part 1: Set Up the Topology and Initialize Devices

Part 2: Configure Devices and Verify Connectivity

- Configure basic settings on PCs, router, and switches.
- Configure OSPF routing on R1, DSW1, and DSW2.

Part 3: Configure and Verify Extended Numbered and Named ACLs

- Configure, apply, and verify a numbered extended ACL.
- Configure, apply, and verify a named extended ACL.

Part 4: Modify and Verify Extended ACLs

Background / Scenario

Extended access control lists (ACLs) are extremely powerful. They offer a much greater degree of control than standard ACLs as to the types of traffic that can be filtered, as well as where the traffic originated and where it is going.

In this lab, you will set up filtering rules for two company branches represented by DSW1 and DSW2. Management has established some access policies between the LANs located at DSW1 and DSW2, which you must implement. The router R1 between DSW1 and DSW2 does not have any ACLs placed on it.

Note: Make sure that the router and switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 1 Router (Cisco 8200L with IOS XE Release 17.09.05a image or comparable)
- 2 Distribution Switches (Cisco 9300L with IOS XE Release 17.09.05a image or comparable)
- 2 Access Switches (Cisco 9200L with IOS XE Release 17.09.05a image or comparable)
- 2 PCs (Windows 10 or above with terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Part 1: Set Up the Topology and Initialize Devices

In Part 1, you will set up the network topology and clear any configurations if necessary.

Step 1: Cable the network as shown in the topology.

Step 2: Initialize and reload the routers and switches.

Part 2: Configure Devices and Verify Connectivity

In Part 2, you will configure basic settings on the router, switches, and PCs. Refer to the Topology and Addressing Table for device names and address information.

Step 1: Configure IP addresses on PC-A and PC-B.

Step 2: Configure basic settings on DSW1.

- Configure the device name as shown in the topology.
- Create a loopback interface on DSW1.
- Configure interface IP addresses as shown in the Topology and Addressing Table.
- Enable web access on DSW1 to simulate a web server with local authentication for user **admin** and password **class**.

```
DSW1(config)# ip http server
DSW1(config)# ip http authentication local
DSW1(config)# username admin privilege 15 secret class
```

Step 3: Configure basic settings on R1.

- Configure the device name as shown in the topology.
- Create the loopback interfaces on R1.
- Configure interface IP addresses as shown in the Topology and Addressing Table.
- Enable web access on R1 following similar configuration as in Step 2d.

Step 4: Configure basic settings on DSW2.

- Configure the device name as shown in the topology.
- Create a loopback interface on DSW2.
- Configure interface IP addresses as shown in the Topology and Addressing Table.
- Enable web access on DSW2 following similar configuration as in Step 2d.
- Enable SSH on DSW2.

```
DSW2(config)# ip domain name cisco.com
DSW2(config)# crypto key generate rsa modulus 2048
DSW2(config)# line vty 0 4
DSW2(config-line)# login local
DSW2(config-line)# transport input ssh
```

Step 5: Configure OSPF routing on R1, DSW1, and DSW2.

- Assign 1 as the OSPF process ID and advertise all networks on R1, DSW1, and DSW2. The OSPF configuration for DSW1 is included for reference.

```
DSW1(config)# router ospf 1
DSW1(config-router)# network 192.168.10.0 0.0.0.255 area 0
DSW1(config-router)# network 192.168.30.0 0.0.0.255 area 0
DSW1(config-router)# network 10.1.1.0 0.0.0.3 area 0
```

- After configuring OSPF on R1, DSW1, and DSW2, verify that the router and distribution switches have complete routing tables listing all networks. Troubleshoot if this is not the case.

Step 6: Verify connectivity between devices.

Note: It is very important to verify connectivity **before** you configure and apply ACLs! Ensure that your network is properly functioning before you start to filter out traffic.

- From DSW1, ping PC-B and the loopback and G1/0/24 interfaces on DSW2. Were your pings successful?
- From DSW2, ping PC-A and the loopback and G1/0/1 interfaces on DSW1. Were your pings successful?
- From PC-A, ping PC-B and the loopback and G1/0/24 interfaces on DSW2. Were your pings successful?
- From PC-B, ping PC-A and the loopback and G1/0/1 interfaces on DSW1. Were your pings successful?
- From PC-A, ping the loopback interfaces on the R1 router. Were your pings successful?
- From PC-B, ping the loopback interfaces on the R1 router. Were your pings successful?

Part 3: Configure and Verify Extended Numbered and Named ACLs

Extended ACLs can filter traffic in many different ways. Extended ACLs can filter on source IP addresses, source ports, destination IP addresses, destination ports, as well as various protocols and services.

Security policies are as follows:

- Allow web traffic originating from the 192.168.10.0/24 network to go to any network.
- Allow an SSH connection from PC-A to DSW2 G1/0/24 interface.
- Allow web traffic originating from the 192.168.20.0/24 network to access DSW1 via the G1/0/1 interface and the 209.165.200.224/27 network on R1. The 192.168.20.0/24 network should NOT be allowed to access any other network via the web.

In looking at the security policies listed above, you will need at least two ACLs to fulfill the security policies. The best practice is to place extended ACLs as close to the source as possible. We will follow this practice for these policies.

Step 1: Configure a numbered extended ACL on DSW1 for security policy numbers 1 and 2.

You will use a numbered extended ACL on DSW1. What are the ranges for extended ACLs?

- Configure the ACL on DSW1. Use 100 for the ACL number.

```
DSW1(config)# access-list 100 remark Allow Web & SSH Access
DSW1(config)# access-list 100 permit tcp host 192.168.10.3 host 10.2.2.1 eq 22
log
DSW1(config)# access-list 100 permit tcp 192.168.10.0 0.0.0.255 any eq 443 log
```

What does the 443 signify in the command output listed above?

To what interface should ACL 100 be applied?

In what direction should ACL 100 be applied?

- b. Apply ACL 100 to the interface.

```
DSW1(config)# interface vlan 1
DSW1(config-if)# ip access-group 100 in
```

- c. Verify ACL 100.

- 1) Open up a web browser on PC-A, and access <https://209.165.200.225> (the R1 router). It should be successful; troubleshoot, if not.

- 2) Establish an SSH connection from PC-A to DSW2 using 10.2.2.1 for the IP address. Log in with **admin** and **class** for your credentials. It should be successful; troubleshoot, if not.

- 3) From privileged EXEC mode prompt on DSW1, issue the **show access-lists** command.

```
DSW1# show access-lists
Extended IP access list 100
10 permit tcp host 192.168.10.3 host 10.2.2.1 eq 22 log (22 matches)
20 permit tcp 192.168.10.0 0.0.0.255 any eq 443 log (111 matches)
```

- 4) From the PC-A command prompt, issue a ping to 10.2.2.1. Explain your results.

Step 2: Configure a named extended ACL on DSW2 for security policy number 3.

- a. Configure the policy on DSW2. Name the ACL WEB-POLICY.

```
DSW2(config)# ip access-list extended WEB-POLICY
DSW2(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 host 10.1.1.1 eq 443
DSW2(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 209.165.200.224
0.0.0.31 eq 443
```

- b. Apply ACL WEB-POLICY to the interface.

```
DSW2(config-ext-nacl)# interface vlan 1
DSW2(config-if)# ip access-group WEB-POLICY in
```

- c. Verify the ACL WEB-POLICY.

- 1) From DSW2 privileged EXEC mode command prompt, issue the **show ip interface vlan 1** command.

What, if any, is the name of the ACL?

In what direction is the ACL applied?

- 2) Open up a web browser on PC-B and access <https://209.165.200.225> (the R1 router). It should be successful; troubleshoot, if not.
- 3) From PC-B, open a web session to <https://10.1.1.1> (DSW1). It should be successful; troubleshoot, if not.
- 4) From PC-B, open a web session to <https://209.165.201.1> (R1 router). It should fail; troubleshoot, if not.
- 5) From PC-B command prompt, ping PC-A. What was your result and why?

Part 4: Modify and Verify Extended ACLs

Because of the ACLs applied on DSW1 and DSW2, no pings or any other kind of traffic is allowed between the LAN networks on DSW1 and DSW2. Management has decided that all traffic between the 192.168.10.0/24 and 192.168.20.0/24 networks should be allowed. You must modify both ACLs on R1 and R3.

Step 1: Modify ACL 100 on DSW1.

- a. From DSW1 privileged EXEC mode, issue the **show access-lists** command.

How many lines are there in this access list?

- b. Enter global configuration mode and modify the ACL on DSW1.

```
DSW1(config)# ip access-list extended 100
DSW1(config-ext-nacl)# 30 permit ip 192.168.10.0 0.0.0.255 192.168.20.0
0.0.0.255
DSW1(config-ext-nacl)# end
```

- c. Issue the **show access-lists** command.

Where did the new line that you have just added appear in ACL 100?

Step 2: Modify ACL WEB-POLICY on DSW2.

- a. From DSW2 privileged EXEC mode, issue the **show access-lists** command.

How many lines are there in this access list?

- b. Enter global configuration mode and modify the ACL on DSW2.

```
DSW2(config)# ip access-list extended WEB-POLICY
DSW2(config-ext-nacl)# 30 permit ip 192.168.20.0 0.0.0.255 192.168.10.0
0.0.0.255
DSW2(config-ext-nacl)# end
```

- c. Issue the **show access-lists** command to verify that the new line was added at the end of the ACL.

Step 3: Verify modified ACLs.

- a. From PC-A, ping the IP address of PC-B. Were the pings successful?
- b. From PC-B, ping the IP address of PC-A. Were the pings successful?

Do you have to apply the ACLs to the interfaces again for the ACLs to work for the pings after you have modified them?

Reflection

1. Why is careful planning and testing of ACLs required?
2. Which type of ACL is better: standard or extended?